

Alexandre-Orso Paoli

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EDUCATION

Columbia University

New York, NY

MSc in Mathematics of Finance (MAFN)

09/2026 – 12/2027

- Coursework: Stochastic Processes, Continuous-Time Finance, Algo-Trading, Numerical Methods, Advanced Machine Learning, Risk Management

Université Paris-Dauphine – PSL

Paris, France

MSc in Applied Mathematics (Financial & Statistical Engineering)

09/2023 – 06/2025

- Coursework: Algorithmic Trading, Structured Products, Interest Rate Models, Credit Risk, Stochastic Mathematics, Model Calibration, Commodity Markets, Monte Carlo Methods, Portfolio Management

BSc in Mathematics and Computer Science

09/2019 – 06/2023

WORK EXPERIENCE

BNP Paribas CIB

Paris, France

Cross-Asset Structurer Intern

10/2025 – 06/2026 (extended)

- Reverse-engineered ~900 structured products (autocalls, repacks, hybrids), €5M–€100M+ notionals, to support desk-wide pricing validation and risk analysis
- Built Python tool automating product reverse-engineering, reducing analysis time from 30 to 5 minutes per product
- Developed LLM-based product analysis and data access tools adopted into the desk's daily production workflow, used by structurers and all incoming interns; presented to 200+ BNP Paribas structurers
- Co-managed Islamic structured products desk (2-person team) as deal flow scaled from €200M to €1Bn
- Contributed to the structuring of a crypto-asset ETF via physical replication through an SPV

Eiffel Investment Group

Paris, France

Investment Analyst Intern

04/2022 – 07/2022

- Analyzed cross-asset correlations and portfolio company exposures to optimize risk-adjusted returns
- Automated risk metrics and financial data processing in Python, saving 3h/week on portfolio monitoring

ACADEMIC PROJECTS

Lead-Lag Analysis – Cross-Asset Macro Markets

Paris, France

Algorithmic Trading Course – Ranked 1st in class

2024 – 2025

- Detected directional lags across 7 macro instruments (FX, rates, commodities) over 3 years of daily data using Dynamic Time Warping and transfer entropy; identified UST13W→UST10Y 4-day and GLD→JPYUSD 1-day lead relationships
- Engineered Optuna objective (MSE + Wasserstein distance directional accuracy) for lag optimization; applied Markov regime-switching (2 regimes) and hierarchical clustering to map cross-asset influence structure

Heston Model Calibration

Paris, France

Model Calibration Course

2024 – 2025

- Calibrated 6-parameter Heston model to 215 live AAPL options (5 maturities, 54 strikes) via SLSQP with L1/L2 Tikhonov regularization; RMSE \$0.42 on mean price \$29.53 (~1.4% relative error)
- Implemented characteristic function inversion with dual numerical integration (rectangular N=10,000 + scipy.quad) and NSS yield curve fitting to US Treasury term structure

Exotic Derivatives Pricer

2024 – 2025

- Live [web app](#) pricing vanilla and exotic options (Asian, Barrier, Lookback, Autocall) via Black-Scholes, Monte Carlo and Binomial Tree, using polynomial fitting for smooth, analytically-derived real-time Greeks

SKILLS & ADDITIONAL INFORMATION

Languages: French (Native), English (C1), Spanish (B1)

Computer Skills: Python (NumPy, pandas, scipy), R, SQL, Bloomberg, Microsoft Office

Extracurricular: President & Co-Founder, Football Club de Boulogne, 2nd division amateur league IDF (2021 – Present)

Interests: Tennis (15 years, competition, referee diploma), Football, Diving (PADI Open Water)